

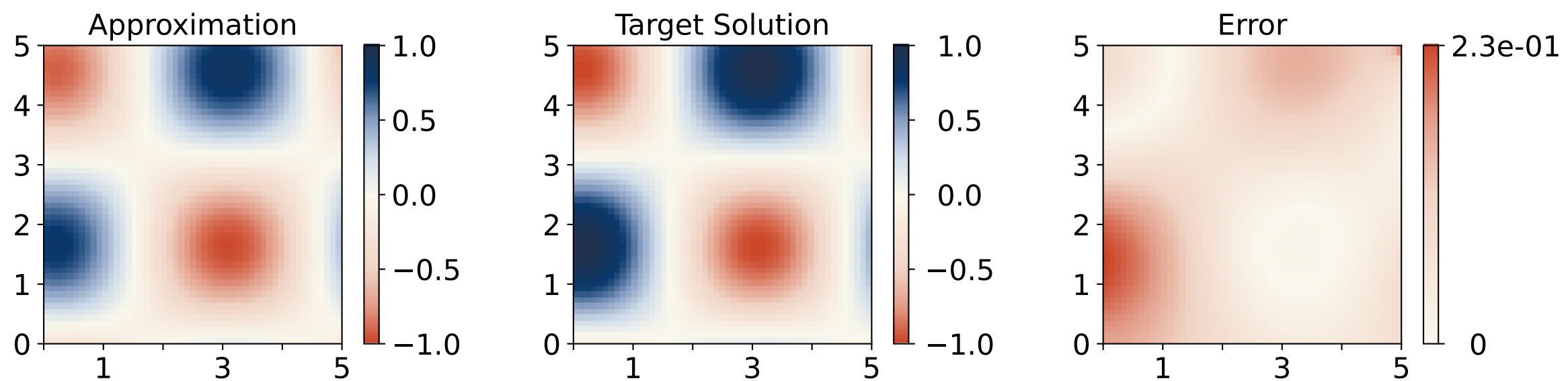


Figure 4.5: Accuracy of Second-Order Finite Difference Approximation of the test function $u = \sin(x)\cos(y)$ with $N = 50$. using known value for u_x , u_y at the boundaries and constraining the boundary norm to be zero.